Bayesian Neural Network based Encrypted Traffic Classification using Initial Handshake Packets

Jiwon Yang, Jargalsaikhan Narantuya, and Hyuk Lim

Wireless Networking and Security (WiSe) Lab.

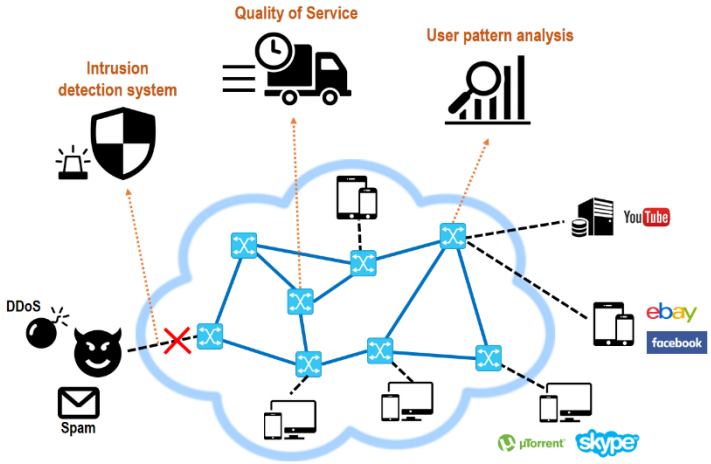
School of Electrical Engineering and Computer Science (EECS)

Gwangju Institute of Science and Technology (GIST)

***Traffic classification* : Identify the data source/destination and categorize the data flow based on some criteria**

- Many researchers try to come up with efficient ways to handle the growing complexity and diversity of internet environment.

- Traffic classification provides solutions : network security, traffic pattern analysis, and quality of service (QoS), etc.

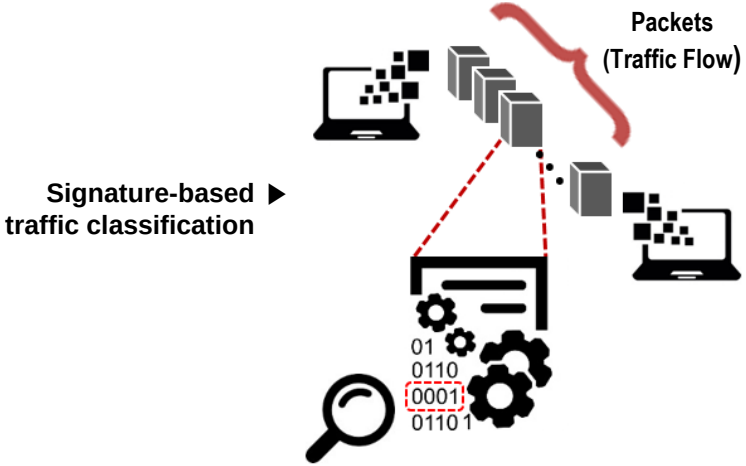


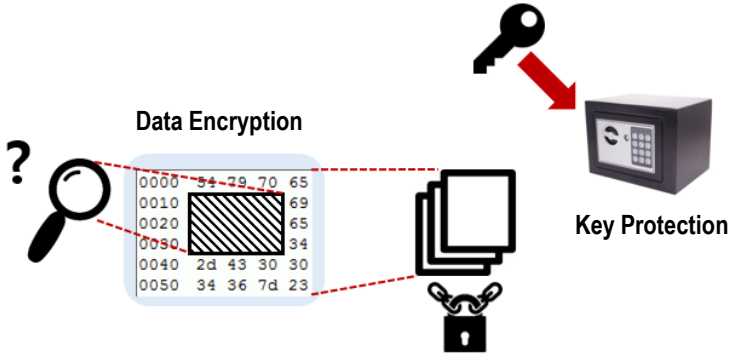
▲ **Several application of network traffic classification**

01 Introduction

Traditional traffic classification approach and challenge

- The traffic classification is performed at application layer level using a small data chunk named packet.
- Most existing works inspect the application-related byte patterns (or signatures) from the packet contents.
- Unfortunately, the signature-based method poses several challenges as the demand for data encryption has been steadily increasing.





▲ The practical challenges for traffic classification

Motivation and our goal

Is there really no information available in traffic flow?
Then, how can two hosts communicate with each other even they had never connected before?

- Handshaking : a process to authenticate peers and allow them to agree on security parameters.
- If we can identify what kind of application data exchanged through the handshaking, we can also classify the following packets in a same traffic flow.
- We propose a novel encrypted traffic classification using initial handshake packets.

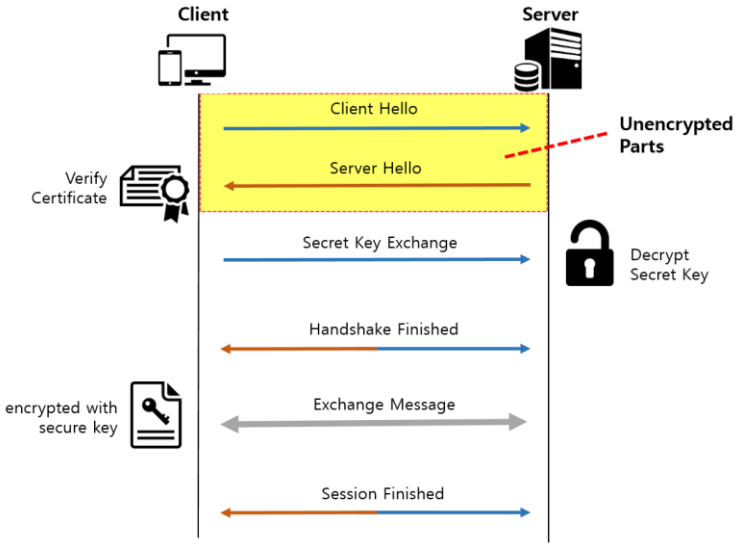
02 Proposed Traffic Classification

Main Target : TLS traffic flows

Transport layer security (TLS) : a popular internet protocols which provide secure communication over the internet

- Ex) HTTPS (for web), SMTPS (for mail service), FTPS (for file transfer), etc.

- We found that the client sends the unencrypted packet named Client Hello as its first message => great potential to contain a lot of useful information



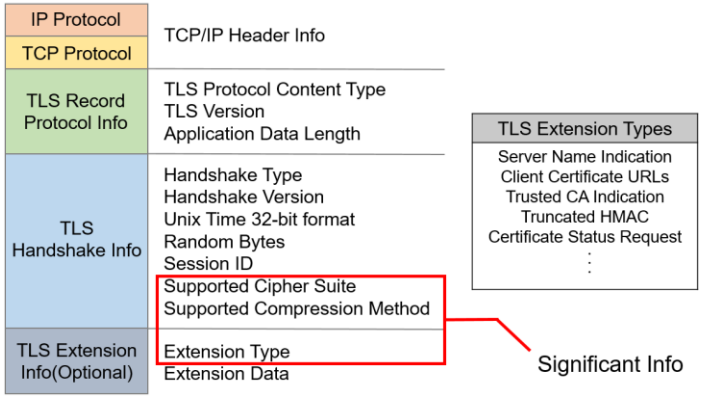
▲ **The detailed steps of TLS communication**

02 Proposed Traffic Classification

Feature selection from TLS handshake packets

Three important contents in TLS client hello message: cipher suite, compression method, and TLS extension

- Cipher suite & Compression method : Each application tends to support its preferred cipher suite and compression method.
- TLS extension : it is designed to provide application-dependent functionalities (e.g. memory limitation)
- In this regard, we select these three contents for traffic classification.



▲ **Contents of TLS Client Hello Messages**

02 Proposed Traffic Classification

Deep learning based classification model

- Checks whether each cipher suite, compression method, and TLS extension is included in a client hello packet.

cipher suites : 344 types compression method : 3 types TLS extension : 45 types

- For each of the total 392 features, mark “1” if the packet has corresponding feature and “0” if it does not.

Extracting features from the packet ▶

cipher suites

```
0000  00 1a 30 f3 46 40 00 16  9c 6f 37 80 08 00 45 00   ··0·F@·· ·o7···E·
0010  00 f9 9e 31 40 00 2e 06  55 c5 7d d4 d9 d6 cb ed   ···1@··· U·}·····
0020  34 70 8c 64 01 bb 96 61  b9 73 d1 e6 2f 6c 50 18   4p·d···a ·s··/lP·
0030  72 10 ab 65 00 00 16 03  01 00 cc 01 00 00 c8 03   r··e···· ........
0040  02 b1 88 e2 7b 4a 2f 0f  ef b8 06 eb 83 19 d1 59   ····{J/· ·......Y
0050  97 05 f7 9e 4e d9 32 31  78 39 13 0d 59 89 2a 3e   ····N·21 x9··Y·*>
0060  fc 00 00 6e c0 14 c0 0a  00 39 00 38 00 37 00 36   ···n···· ·9·8·7·6
0070  00 88 00 87 00 86 00 85  c0 0f c0 05 00 35 00 84   ........ ·····5··
0080  c0 13 c0 09 00 33 00 32  00 31 00 30 00 9a 00 99   ·····3·2 ·1·0····
0090  00 98 00 97 00 45 00 44  00 43 00 42 c0 0e c0 04   ·····E·D ·C·B····
00a0  00 2f 00 96 00 41 00 07  c0 11 c0 07 c0 0c c0 02   ·/···A·· ........
00b0  00 05 00 04 c0 12 c0 08  00 16 00 13 00 10 00 0d   ........ ........
00c0  c0 0d c0 03 00 0a 00 15  00 12 00 0f 00 0c 00 09   ........ ........
00d0  00 ff 01 00 00 31 00 0b  00 04 03 00 01 02 00 0a   ·····1·· ........
00e0  00 1c 00 1a 00 17 00 19  00 1c 00 1b 00 18 00 1a   ........ ........
00f0  00 16 00 0e 00 0d 00 0b  00 0c 00 09 00 0a 00 23   ........ .......#
0100  00 00 00 0f 00 01 01                              .......
```

compression method TLS extension

Ex) Cipher #60 = RSA_with_AES_128_CBC_SHA256
Extension #1 = Max_Fragment_Length

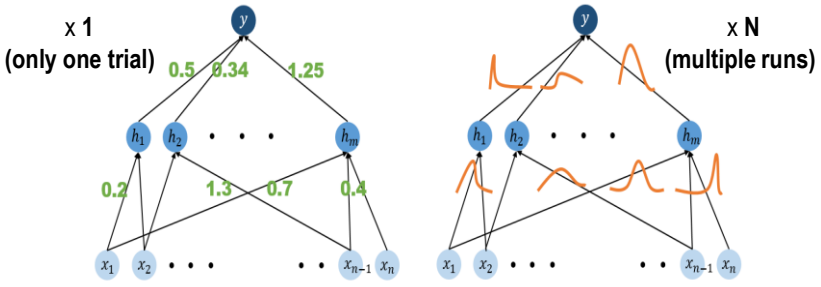
0 : not supported, 1 : supported

Traffic	Cipher A	Cipher B	...	Compression A	Compression B	...	Extension A	Extension B	...
Packet 1	1	0		0	0		1	0	
Packet 2	0	1		1	0		1	0	
...									

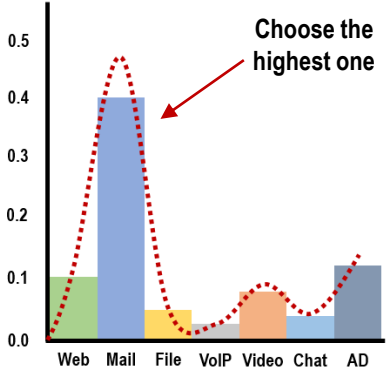
▲ **Vector conversion of supported features**

Bayesian Neural Network (BNN) : designed to figure out the uncertainty which can be caused by the unfamiliar data

- Do multiple runs of the BNN to classify input data (each time with a new set of sampled weight)
- This model is appropriate to our classification because it helps to discover the weak point of the neural network model and estimate uncertainty.



▲ General Neural Net (left) and Bayesian Neural Net (right)



◄ Probability density function of the set of outputs

Dataset & Experiments

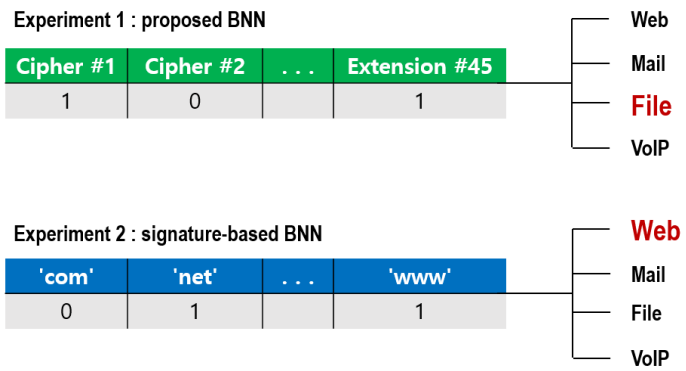
- Total of 328,155 traffic data with four categories (web, mail, file, and VoIP).

- Experiment 1 : proposed idea
  BNN with List of cipher suite/compression method/extension

- Experiment 2 : signature inspection
  BNN with 100 types of pre-selected byte patterns
  (e.g., ‘com’, ‘net’, ‘www’, ‘mail’, 'file', 'youtube', etc)

Protocol	Size(KB)	# of flows	Rate(%)	Content
Web	98,366	83,353	25.4	HTTPS
Mail	63,441	82,128	25.0	SMTPS, POP3S, IMAPS
File	90,988	80,095	24.4	FTPS
VoIP	114,895	82,579	25.2	XMPP over TLS

▲ **Selected TLS traffic data set**



▲ **Comparative Experiment**

Results

Experiment 1)

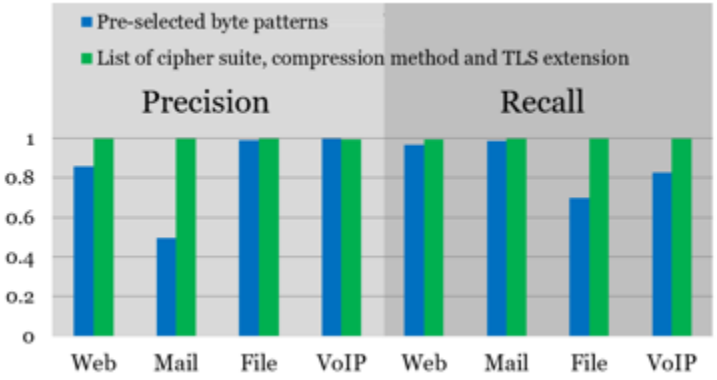
Proposed BNN (Green) using the cipher suites, compression method and TLS extension shows an excellent performance

=> It achieves over 0.99 of precision and recall for every category.

Experiment 2)

Signature-based BNN (Blue) obtains unbalanced results.

⇒ bad training : except for web traffic, many data do not have any signature-related information.

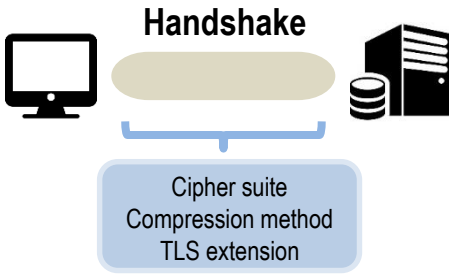


▲ **Precision and recall results of the classification methods**

04 Conclusion

Conclusion & Future work

- We proposed a novel traffic classification method for secure communication.
- Cipher suites, compression methods, and extension is the key features for TLS traffic classification.
- The experiment results showed that the proposed method performed better than the traditional signature-based classification method.
- we plan to extend our work to other secure protocols such as IPsec and SSH.



Thank You.